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Multistage semisupervised active learning framework for crack identification, segmentation, and measurement of bridges

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Abstract

In bridge health monitoring (BHM), crack identification and width measurement are two of the most important indices for evaluating the functionality of bridges. In order to reduce the labor cost in field detection, researchers have proposed a variety of deep learning (DL)-based detection techniques for crack recognition. However, some problems still exist in extending these techniques to practical applications, such as data annotation difficulty, limited model generalization ability, and inaccuracy of the DL identification of the actual crack width measurement. In this paper, an application-oriented multistage crack recognition framework is proposed, namely, Convolutional Active Learning Identification-Segmentation-Measurement (CAL-ISM). It includes four steps: (1) pretraining of the benchmark classification model, (2) retraining of the semisupervised active learning model, (3) pixel-level crack segmentation, and (4) crack width measurement. Beyond numerical experiments, the performance of the CAL-ISM is validated for practical applications: (i) bridge column test specimen and (ii) field BHM projects. In conclusion, the obtained results from these applications shed light on the high potential of CAL-ISM for BHM applications, which is recommended in future deployments for BHM.

1 | BACKGROUND AND MOTIVATIONS

By the end of 2020, there are 912,800 highway bridges in China, including 6444 long-span bridges and 119,935 normal bridges. The total mileage of the highway is about 5.19 million kilometers (km), of which the maintenance mileage of the highway is 5.14 million km, accounting for 99.0% of the total mileage (Ministry of transport of the People's Republic of China, 2020). The remaining 1% is for newly constructed highways with no need for immediate maintenance. Similarly, there are more than 618,000 bridges in the United States, where nearly 36% of the bridges need repair work, and 7.3% of them are considered structurally deficient (American Road and Transportation Builders Association, 2021). Therefore, surging demands in the maintenance work of highway bridges in China, the United States, and many other countries are worldwide realities and it is a necessity to develop effective methods to evaluate the service functionalities of bridge structures. Because of cost-efficiency and easy-forming, reinforced

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concrete (RC) is used for most of bridge structures worldwide. However, damage to RC is commonly initiating on its surface in the form of cracks. The appearance of cracks not only affects the bearing capacity of the structure, but also leads to possible corrosion of internal reinforcement, which greatly reduces the service life of the structure. Therefore, accurate identification and measurement of these cracks are of great importance during bridge health monitoring (BHM).

In the practice of BHM, on-site manual inspection is most commonly adopted. Inspectors and engineers may utilize inspection vehicles, equipped with a variety of detection instruments and forms for data recording, to inspect all parts of the bridges and recommend appropriate maintenance actions. However, human inspection is typically laborious and subjective, resulting in possible erroneous measurements and decisions. In addition, using inspection vehicles often requires inspectors and engineers to reach the intended position for inspection through platforms and end up working at high elevations above the ground, which brings about safety concerns. In recent years, optical fiber sensing technology has been developed for BHM, which can perform tasks such as crack width determination, positioning, and orientation for large systems such as long-span bridges. However, due to the complex layout, need for skillful labor, and high cost due to expensive equipment and sensors, it is not widely used. In the past few decades, computer vision (CV) technologies provide another direction for BHM, and many traditional CV methods, for example, histogram transformation (Otsu, 1979), threshold segmentation (Cheng et al., 2003; Fujita et al., 2006; Otsu, 1979), edge detection (Behera et al., 2012; Martin et al., 2004), and seed region growth (Huang & Xu, 2006; Li et al., 2011), have been explored in the vision-based crack detection. However, the detection accuracy of these methods does not meet the engineering requirements for practical decisions.

Nowadays, machine learning (ML) and deep learning (DL) have been at the forefront of a plethora of research activities in many disciplines, where topics in structural engineering, for example, system parameter identification (Perez-Ramirez et al., 2016), bolt loosening detection (Yang et al., n.d.), and crack detection, significantly benefit from these technologies. In order to reduce the cost and safety risk while improving detection accuracy and automation, researchers have proposed a variety of image detection technologies combining ML/DL with CV to recognize cracks (Kong et al., 2021; Liang, 2019; Miao et al., 2021; Żarski et al., 2021; Zhang & Yuen, 2021). In general, vision-based crack detection has three main tasks: identification, localization, and segmentation. Cha et al. (2017) built a convolutional neural network (CNN) for detecting concrete cracks without calculating the defect features. Their results demonstrated that this CNN achieved better performance than the traditional Canny and Sobel edge detection methods (Canny, 1986; Kanopoulos et al., 1988). Dung and Anh (2018) proposed a method to conduct cutting tasks on surface crack images of concrete based on deep fully CNN to separate the cracks from the images and to identify their locations in the images. However, the method is shown to be susceptible to ambient noise. Ju et al. (2019) developed a crack deep network based on the fast region CNN, which is robust in crack detection but requires intense computational resources. Liu et al. (2019) applied Adam algorithm to achieve fast convergence of the parameter iterations based on U-Net showing higher accuracy only with smaller training set. Liu et al. (2020) proposed a two-step crack detection and segmentation method based on CNN where the modified "You Only Look Once" (YOLO-V3) and U-Net were, respectively, used for these two tasks. Finally, a pixel-level detection method was proposed to identify cracks based on a deep convolutional encoder–decoder network (Bang et al., 2019).

Most of the above-mentioned studies fall within the supervised learning, which requires data–label pairs for model training. However, there are several concerns in supervised learning, such as the difficulty of obtaining a large amount of labeled data and the limited generalization ability (the model's ability to adapt properly to new, previously unseen data). Therefore, it is of great engineering significance to utilize unlabeled data, which are easily obtained, to improve the role of supervised models (Gao et al., 2021). An algorithm that combined an adversarial learning method with a full convolution discriminator was proposed to generate a supervisory signal for unlabeled images to make up for the deficiency of image annotation (Li et al., 2020). In this way, semisupervised learning could provide an alternative solution using both labeled and unlabeled data to train the model. Many studies on semisupervised learning have dealt with data lacking annotation. Wang and Su (2021) proposed a semisupervised semantic segmentation network for surface crack detection based on Efficient U-Net to significantly reduce the workload of annotation while maintaining high accuracy. Di et al. (2019) combined the convolutional autoencoder (CAE) and semisupervised generative adversarial network (SGAN) to detect surface defects of steel members. The study revealed that the CAE–SGAN method can automatically extract deep characteristics from images with better performance than conventional methods. Li et al. (2021) deeply combined dense atrous convolution (DAC) block and residual multikernel pooling (RMP) block to detect cracks of bridges at pixel level showing excellent extraction capability of information and features from images for a case study. Gao et al. (2021) proposed a balanced semisupervised GAN (BSS-GAN) method by

adopting semisupervised learning and balanced-batch sampling technique in training to jointly and effectively resolve the low-data and imbalanced-class classification problems.

Once the crack has been identified, accurately measuring its width from vision data is the next important task. For the past several years, many studies have been conducted to develop methods for vision/image-based crack width measurements. Jahanshahi et al. (2013) introduced a contactless crack quantification approach to estimate the crack width, which was computed by the orientation (perpendicular to the crack centerline) at each centerline pixel. Subsequently, Jahanshahi and Masri (2013) proposed a fully autonomous, fully contactless crack quantification approach. In this approach, the authors used morphological features and a set of strip kernels to compute the effective crack width. Lu et al. (2016) presented a double-threshold algorithm to identify the width of both fine and wide cracks. Li et al. (2017) developed a fully automatic ML-based algorithm, which combined a modified region-based active contour model for image segmentation and a linear support vector machine for noise elimination. Ni et al. (2019) proposed a DL-enabled quantitative crack width measurement method based on dual-scale CNN. They also developed a crack width estimation method based on Zernike moment operator (Ghosal & Mehrotra, 1993) for thin cracks.

Although tremendous effort has been made on vision-based crack detection for damaged structures, much of this effort was either based on the ideal conditions with well-processed data or focused on a single topic, for example, crack classification or crack segmentation. There still exists a gap between the developed vision/image-based technologies and their applications in real-world projects. In this paper, to unify the procedures of crack identification, segmentation, and width measurement along with better recognition performance for the practical scenarios (less labeled data annotated by human experts and more unlabeled data, which are only images or videos without annotations), we propose an integrated multistep framework adopting semisupervised learning, namely, Convolutional Active Learning Identification-Segmentation-Measurement (CAL-ISM) with four steps: (1) benchmark classification model pretraining, (2) semisupervised active learning model retraining, (3) pixel-level crack segmentation, and (4) crack width measurement. The performance of the proposed CAL-ISM framework is validated by data collected from practical applications: (1) bridge column test specimen and (2) bridge structure field monitoring projects, as an example for future BHM projects.

Specific technical contributions are summarized below:

(1) A semisupervised active learning method is proposed for a recognition model training using labeled and unlabeled data simultaneously to reduce labeling demands compared to conventional active learning methods.
(2) A stacked convolutional autoencoder (SCAE) structure updated by unlabeled data is adopted to extract data features without labeling information to be combined with labeled data features via feature fusion techniques to improve the model recognition performance.
(3) A modified Sobel operator method is developed along with an 8-direction region growing algorithm to process the masks generated by the segmentation model, and subsequently to extract refined crack edges for crack width measurements linking DL approaches and conventional crack analysis methods.
(4) Candidate model mechanism is applied to select the best recognition and segmentation models for a BHM project, which improves both accuracy and robustness of the BHM process.

2 | SEMISUPERVISED ACTIVE LEARNING

Because manually labeling samples is time-consuming, it is usually difficult to obtain a large amount of labeled data compared to having unlabeled data. If the semisupervised learning method is employed, both labeled and unlabeled data can be utilized simultaneously. Active learning is another technique to complement the semisupervised learning. In active learning, a group of unlabeled data, which are recognized by the classifier with low confidence level (low probability of predicting a specific class), are first queried. Subsequently, these data are labeled by human annotators, and then fed into the original classifier for performance improvement. In addition, autoencoder (AE)-type network, for example, SCAE (Masci et al., 2011), is known as a good feature extractor in unlabeled data, which helps the model better utilize the unlabeled data (Vincent et al., 2010). In this study, we combine semisupervised learning, active learning, and SCAE, into a new training paradigm, that is, SCAE-based active learning.

2.1 | Self-training

Self-training (Clark et al., 2003) is one early realization of the active learning. It does not rely on complex assumptions and its algorithm structure is straightforward. For a K-class classification problem, it is supposed that there are N_l pairs of labeled data $\{(X^l, Y^l)\}$, and N_{ul} *initial*

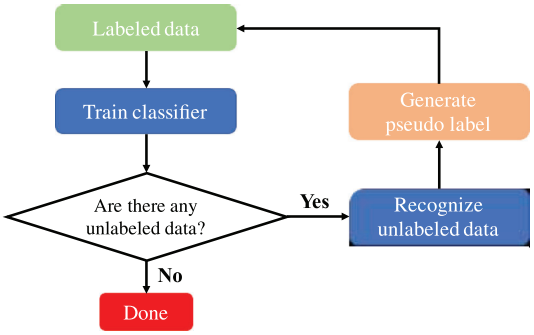


FIGURE 1 Procedure of self-training

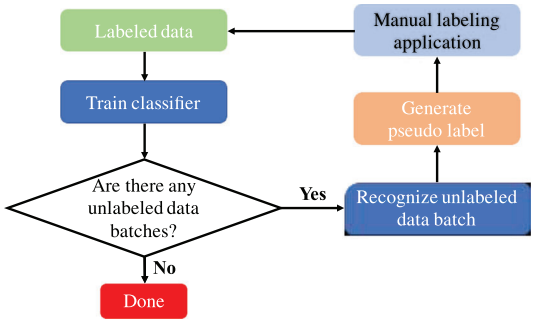


FIGURE 2 Procedure of active training

unlabeled data $\{X^{ul}\}$, where X represents the data or the data features in a general way, and Y represents the labels. An *initial* training set T is formed with all labeled data, $T = \{(X^l, Y^l)\}$. The related algorithm (Figure 1) is as follows:

1. Train a classifier with all labeled data in T.
2. Use the trained classifier to predict the unlabeled data set $\{X^{ul}\}$, and generate the K-dimension output $\{O^{ul}\}$.
3. Select N_{hc} images with high confidence, $\{X^{hc}\}$, from $\{X^{ul}\}$, and assign them the pseudo-label $\{O^{hc}\}$ based on their outputs, $\{O^{ul}\}$, from step 2.
4. Remove $\{X^{hc}\}$ from the unlabeled set $\{X^{ul}\}$, and add them to *update* the training set T and obtain:

$$T \leftarrow \{(X^l, Y^l)\} \bigcup \{(X^{hc}, O^{hc})\} \tag{1}$$

$$\{X^{ul}\} \leftarrow \{X^{ul}\} - \{X^{hc}\} \tag{2}$$

5. Retrain the classifier with the updated training set T.
6. Repeat steps 2 to 5 until all unlabeled data are labeled.

It is noted that in step 3, a threshold ϵ_{cl} for filtering data with confidence level is predefined. If the probability of predicting a specific class is higher than ϵ_{cl}, the input data are defined as having high confidence, and vice versa. The value of ϵ_{cl} is defined by the user based on demands, for example, the higher the value of ϵ_{cl}, the stricter the criterion for data filtering, and the more manual labeling work is expected.

2.2 | Active learning

Although self-training can somewhat improve the baseline performance utilizing new unlabeled data, the improvement may be limited because the pretrained model has limited recognition ability leading to labeling errors (Settles, 2009), that is, generated pseudo-label with high confidence is incorrect. These errors make the model unable to update its parameters, and may even lead to overfitting, and ultimately deteriorating the model performance. Therefore, if manual labeling is conducted on those hard-to-distinguish images, the model can correct the wrong parameters and improve its recognition performance. Active learning plays such a role. For example, in the crack detection problem, if the confidence levels of predicting "non-cracked" (none of the cracks occur on the surface of the structural component) and "cracked" (cracks occur on the surface of the structural component) are close for images recognized by the model, it is inferred that it is hard to recognize the real conditions and distinguish the classes by this model. Under such circumstances, those images should be subjected to manual labeling.

Based on the defined K-class classification in Section 2.1, the procedure of the active learning (Figure 2) is as follows:

1. Train a classifier with all labeled data in T.
2. Randomly separate the initial unlabeled data set $\{X^{ul}\}$ to M batches, that is, $\{X^{ul}\}_1, \ldots, \{X^{ul}\}_M$.
3. Randomly select one batch $\{X^{ul}\}_m$, $m \in \{1, \ldots, M\}$, use the trained classifier to predict the labels for all unlabeled data, and then select the N_d most difficult images $\{X^{rl}\}$ with low confidence levels for manual labeling. These N_d images obtain their real labels manually, and form the data–label pairs as $\{(X^{rl}, Y^{rl})\}$.
4. To increase the number of $\{(X^{rl}, Y^{rl})\}$, off-line data augmentation (Gao & Mosalam, 2018) can be conducted to generate more data–label pairs, resulting in a larger $\{(\hat{X}^{rl}, \hat{Y}^{rl})\}$.
5. Remove the selected batch $\{X^{ul}\}_m$ from the unlabeled data set $\{X^{ul}\}$, and add $\{(\hat{X}^{rl}, \hat{Y}^{rl})\}$ to *update* the labeled data set T as follows:

$$T \leftarrow \{(X^l, Y^l)\} \bigcup \{(\hat{X}^{rl}, \hat{Y}^{rl})\} \tag{3}$$

$$\{X^{ul}\} \leftarrow \{X^{ul}\} - \{X^{ul}\}_m \tag{4}$$

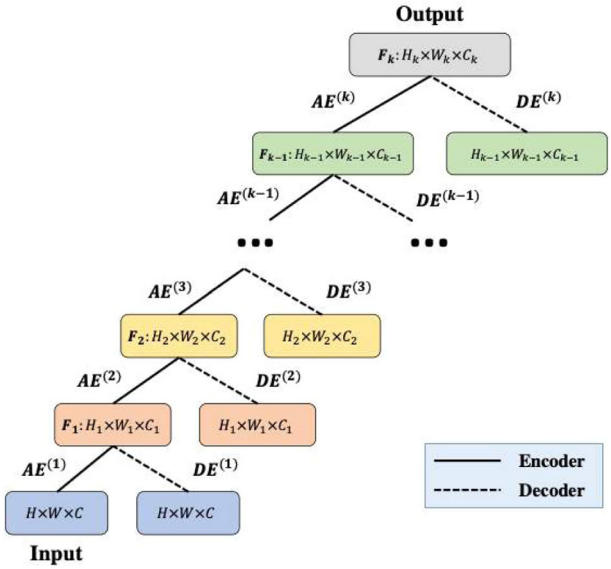


FIGURE 3 Illustration of stacking structure of the SCAE

6. Retrain the classifier with the updated data set T.
7. Repeat steps 3 to 6 until all M unlabeled batches are removed from $\{X^{ul}\}$.

2.3 | SCAE-based active learning

In order to further utilize informative features in the data while conducting active learning, SCAE is adopted into the active learning loop. The SCAE is one type of CAE structure, where several CAE's can be stacked to form a deep hierarchy (Masci et al., 2011). As illustrated in Figure 3, an SCAE consists of several layers of CAE, where the output of each layer is connected to the input of the successive layer. Through nonlinear dimensional reduction, high-dimensional data are mapped to the low-dimensional space, and thus the hidden features are extracted. No labeling information is required, so SCAE can utilize abundant unlabeled data to obtain a good feature extraction ability. Subsequently, through feature fusion, the features extracted by SCAE from the data without considering labeling information are combined with those extracted by the original classifier from the labeled data.

Suppose the input image I is $(H \times W \times C)$, where H, W, and C are its height, width, and number of color channels, respectively. The first AE layer, $AE^{(1)}$, generates the feature F_1 with the shape (H_1, W_1, C_1), and the output of the corresponding decoder (DE) layer, $DE^{(1)}$, has the same shape of the input I. Subsequently, using F_1 as the input to the second AE layer, $AE^{(2)}$, second layer feature F_2 is extracted with the shape (H_2, W_2, C_2), and the second DE layer, $DE^{(2)}$, maps F_2 back to have the shape of F_1. Keep stacking until the designated depth k, and the final feature F_k is extracted by $AE^{(k)}$ with the shape (H_k, W_k, C_k). It is noted that while designing the network structure of the SCAE, the final feature F_k should share the same dimension with the features F_{clf} extracted by the original classifier (step 1 in the active learning procedure), which is expected to be a neural network–like model.

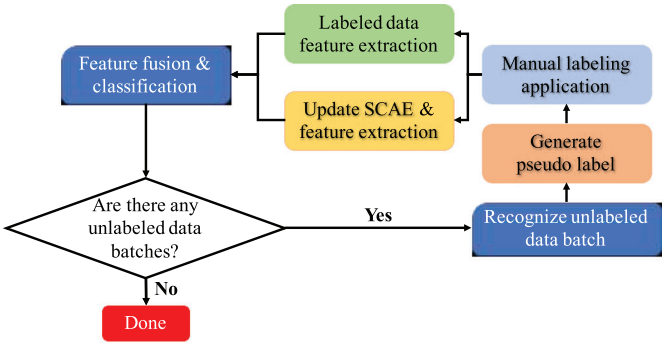


FIGURE 4 Procedure of SCAE-based active training

One of the basic assumptions of the *feature fusion* is that different feature extraction methods have different representative abilities. Through a parallel combination of multiple features from different sources, a relatively comprehensive feature group can be obtained, where features extracted by different methods can complement each other. Therefore, after feature fusion, the performance of the model is expected to be improved. In this study, SCAE with a feature fusion technique is applied along with active learning. Figure 4 illustrates that the updated labeled data set T (step 5 in the active learning procedure) is used for feature fusion of the features F_{clf}, which are extracted by the original classifier using labeling information and features F_k, which are, on the other hand, extracted by the SCAE without labeling information. This is a *semi-supervised learning*. Both F_{clf} and F_k are concatenated to the shape $(H_k, W_k, 2C_k)$ first, and then Global Pooling (Goodfellow et al., 2016) is applied to reduce its dimension to a $(1 \times 1 \times C_k)$ feature vector F_{fu}. Subsequently, F_{fu} is fully connected (FC) to the output layer with the number of neurons that is equal to the number of total classes. The procedure is summarized as follows (note that the first five steps are identical to those in the procedure shown in Section 2.2):

1. Update SCAE using unlabeled data $\{X^{ul}\}_m$, and extract the features F_k from the updated labeled data set T.
2. Retrain the classifier with the updated labeled data set T, and extract the features F_{clf}.
3. Conduct feature fusion for F_k and F_{clf} to generate the fusion feature F_{fu}.
4. Train the single-layer FC network with F_{fu} as the input to this FC network.

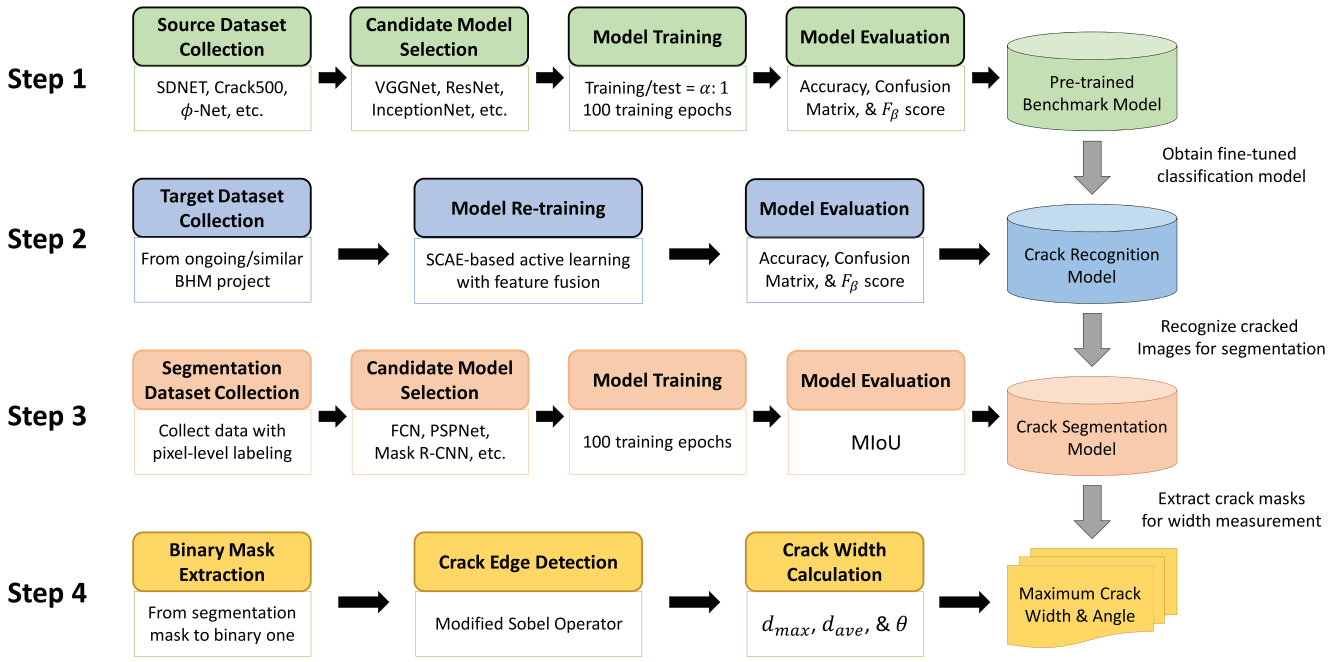


FIGURE 5 CAL-ISM framework

5. Repeat steps 3 to 9 until all unlabeled batches are removed from $\{X^{ul}\}$.

3 | PROPOSED CAL-ISM FRAMEWORK

To link the gap between vision-based technologies and their applications in real-world projects, the integrated multistep framework CAL-ISM is proposed here as illustrated in Figure 5. Based on findings from previous studies: (i) the training uncertainty can be reduced by adopting a transfer learning technique (Gao & Mosalam, 2018, 2020), that is, retraining a benchmark model through tuning its parameters has a better and more stable performance than training directly from scratch, and (ii) the usage of the unlabeled images can reduce the labeling efforts and improve the model generalization ability (Gao et al., 2021; Li et al., 2020; Wang & Su, 2021). The working principle of the CAL-ISM framework is organized into four major steps: (1) pretraining benchmark classification model, (2) retraining semisupervised active learning model, (3) pixel-level crack segmentation, and (4) crack width measurement. The details of these four steps are described in the following subsections.

3.1 | Step 1: Pretraining benchmark classification model

The objective of this step is to determine a pretrained benchmark model, which can be used in a general crack identification (classification) problem, and then can be easily adjusted based on newly collected data. First, a pretraining data set is built, where data can be collected from several open-source structural/crack data sets, for example, SDNET (Dorafshan et al., 2018), Crack500 (Zhang et al., 2016), and ϕ-Net (Gao & Mosalam, 2020). Then, the data are split into training and test sets with a ratio of α : 1 for the purpose of model performance evaluation. A variety of current mainstream networks, for example, VGGNet (Simonyan & Zisserman, 2014), ResNet (He et al., 2016), and Inception-ResNet (Szegedy et al., 2017), are selected as the candidates of the benchmark model. Subsequently, these models are trained and tested based on the pretraining data and their performance is evaluated through multiple metrics, that is, accuracy, confusion matrix (CM), and F_β score (Baeza-Yates & Ribeiro-Neto, 1999).

The accuracy is defined by the number of correct predictions divided by the total number of predictions made for a data set. It is one of the mostly used metrics in classification. However, in the practice of BHM, the occurrences of damage, for example, "cracked" are far less than "non-cracked." Thus, the model having a high accuracy may predict all the inputs as "non-cracked" resulting in a biased model. Thus, CM is also adopted here. For a binary classification problem, denote "cracked" as a positive class and "non-cracked" as a negative class, and the CM has four possible outcomes: true positive (TP), true negative (TN), false positive (FP), and false negative (FN). For example, TP represents the scenario that the model predicts the input as the positive class "cracked,"

which is consistent with the ground truth "cracked." Based on these outcomes, *recall* also referred to as the *true positive rate* (TPR) and *precision*, defined in Equation (5), and *true negative rate* (TNR), defined in Equation (6), are computed. It is noted that recall is the proportion of images with cracks that can be recognized by the model, which reflects the comprehensiveness of recognition, and precision is the proportion of images with cracks in the predicted images, which reflects the accuracy of recognition. In order to take into account both recall and precision of the model simultaneously, F_β score is adopted. For any β value, the F_β score is calculated by Equation (7). Finally, according to this comprehensive model performance evaluation, the relatively optimal model with highest F_β score or least parameters if F_β scores are close (within 1%) is selected as the benchmark model.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad \& \quad \text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{5}$$

$$\text{TNR} = \frac{\text{TN}}{\text{TN} + \text{FP}} \tag{6}$$

$$F_\beta = \left(1 + \beta^2\right) \times \frac{\text{Recall} \times \text{Precision}}{\text{Recall} + (\beta^2 \times \text{Precision})} \tag{7}$$

3.2 | Step 2: Retraining semisupervised active learning model

In this step, the techniques of transfer learning and domain adaptation are adopted to transfer information and knowledge from the source domain to the target domain (Gao & Mosalam, 2018; Pan & Yang, 2009). Specifically, the benchmark model pretrained on the source data set in step 1 is treated as a basis of learning general damage due to cracking, and it is further retrained on a custom target data set to identify the damage on the specific bridge structure and components, where data are acquired from the bridge structure for the BHM project at hand or from a similar scenario to this specific project.

Due to costly manual labeling, the custom target data set usually consists of two parts: a small portion of labeled data and a larger portion of unlabeled data. The traditional supervised learning methods only work on the labeled data, neglecting the abundant information of unlabeled data. However, here, SCAE-based active learning with feature fusion is used, which is a semisupervised method to utilize information from both labeled and unlabeled data simultaneously. It combines the features from the labeled data through the original classifier and from the unlabeled data through SCAE, where the ratio between unlabeled and labeled data is set as $\gamma : 1$. Through the active learning mechanism, a small portion of the unlabeled data, initially recognized by the classifier with less confidence, is subsequently manually labeled and added to the labeled data set for improving the classifier performance. In the classification problem of step 2, the model evaluation metrics are the same as in step 1, that is, accuracy, CM, and F_β score. Finally, a well-tuned recognition model based on the custom data set is obtained and directly used for the crack identification task as the intended target of the BHM project at hand. The "cracked" images recognized by this model are further investigated using a crack segmentation model in step 3.

3.3 | Step 3: Pixel-level crack segmentation

Since the occurrence of the crack is identified in step 2, the pixel region of the crack is expected to be recognized in step 3. In this step, through feeding the "cracked" images, recognized by the updated benchmark model in step 2, the segmentation model generates the predictive crack pixel regions, which are further used for crack width measurement in step 4.

Similar to step 1, the final segmentation model is selected from multiple candidates, where three classic models, namely, fully convolutional network (FCN) (Long et al., 2015), Pyramid Scene Parsing Network (PSPNet) (Zhao et al., 2017), and Mask region–based CNN (Mask R-CNN or MR-CNN for short) (He et al., 2017), are adopted in this study. FCN recognizes the class of each image pixel based on highly abstract features. Through "skip structure" (He et al., 2016), the shallow and deep features are connected, where the former usually represents the fine-scale features, and the latter reflects the semantic information. Thus, FCN model combines both fine-scale features and abstract semantic information, leading to more accurate performance for local parts of the image, which is useful for crack segmentation. To understand complex scenarios and avoid confusion between targets, PSPNet was proposed using multiple receptive fields of different sizes. In this way, the network uses global information when analyzing the local information, and the probability of false segmentation is reduced. Nowadays, one of the most efficient semantic segmentation model is a two-stage framework, namely, MR-CNN. In the first stage, through feature extraction network, it extracts multilevel features of the input image. Subsequently, these features are fed into a region proposal network (RPN) by sliding windows to obtain the region of interest (RoI). In the second stage, binary masks are generated for each RoI, which contribute to the final mask image for the crack.

It is noted that ML/DL performance usually varies for different data sets but using different candidate models leads to more robust and accurate results.

Since the main outcome of conducting segmentation is to generate a mask image, the evaluation metric is usually based on the comparison between the mask image and the ground truth pixel-level labeling. In this study, the pixel-level label only contains binary information, that is, the occurrence of the crack is 1, otherwise 0. Thus, a specific method for segmentation model evaluation, namely, Mask Intersection over Union (MIoU) (He et al., 2017), is used. It has the advantages of nonnegativity, symmetry, and scale invariance. Denoting the recognized image as I, where (x, y) represents an arbitrary pixel, P, the mask is generated by the segmentation model as M_c, and the ground truth pixel-level is labeled as M_g. Define $\mathbb{1}\{\cdot\}$ as the indicator function, that is, if the inside condition is true, it is 1, otherwise 0. Thus, MIoU is computed as follows:

$$\text{Inter} = \sum_{(x,y)\in P} \mathbb{1}\{M_c(x,y) + M_g(x,y) = 2\} \tag{8}$$

$$\text{Union} = \sum_{(x,y)\in P} \left[\mathbb{1}\{M_c(x,y) = 1\} + \mathbb{1}\{M_g(x,y) = 1\}\right] - \text{Inter} \tag{9}$$

$$\text{MIoU} = \text{Inter} \Big/ \text{Union} \tag{10}$$

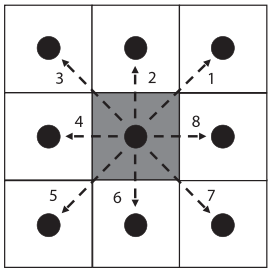


FIGURE 6 The 8-direction region growth

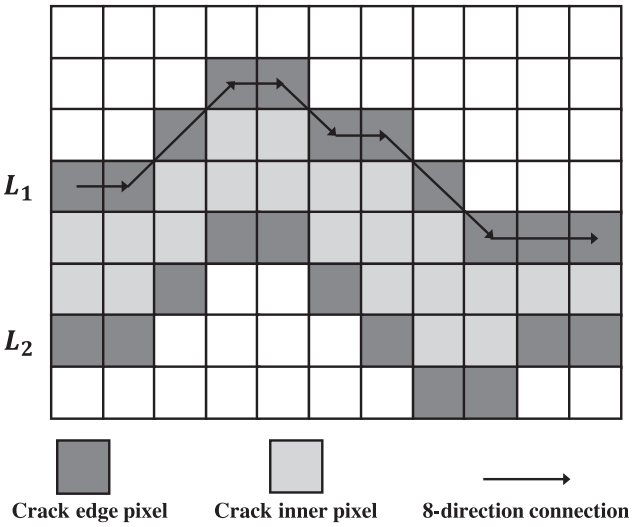


FIGURE 7 The 8-direction crack edge extraction

3.4 | Step 4: Crack width measurement

From step 3, a binary mask of the crack image is obtained, where the foreground is the crack pixels with label "1" and the background is intact concrete surface pixels with label "0." For the purpose of measuring the crack width, two edges of the crack need to be found in the mask image. In this framework, one variant of Sobel operator (Kanopoulos et al., 1988) (denoted as modified Sobel operator) is proposed. Its basic assumption is that the pixel value changes significantly near the edge, and such a change can be represented by the gradient of the image. Given one image, the operator uses two 3×3 kernels, which are convolved with the original image to calculate approximations of the gradients. Here, G_x and G_y represent two kernels, which at each point contain the horizontal and vertical gradient approximations, respectively. Subsequently, the approximate gradient G is as follows:

$$G = \sqrt{{G_x}^2 + {G_y}^2} \tag{11}$$

The larger the value of G at a point, the more likely the point is an edge, which can either be a crack or a background. However, the extracted edge by the conventional Sobel operator (Kanopoulos et al., 1988) may not be precise as it may contain a large number of nearby pixels. To refine these edges, a modified gradient, G_{mod}, is computed by multiplying G with the pixel value f at any point, that is,

$$G_{mod} = G \cdot f \tag{12}$$

It is noted that here the input image is the binary mask, which has only binary pixel values, that is, 0 or 1.

Computing each point in the mask, two edges of one crack (L_1 & L_2) can be found through an 8-direction region growing algorithm (Tsai, 1987), Figures 6 and 7. Based on the extracted edges of the mask, any pixel on L_1 or L_2 is replaced by the corner of the outer edge of the pixel, for example, (x_i, y_i) on L_1 and (x_j, y_j) on L_2, Figure 8. Subsequently, for an arbitrary point (x_i, y_i), its minimal Euclidean distance to (x_j, y_j) is computed by Equation 13, where $\langle\cdot\rangle$ represents the Euclidean distance computation between two points, that is, $\sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$. The

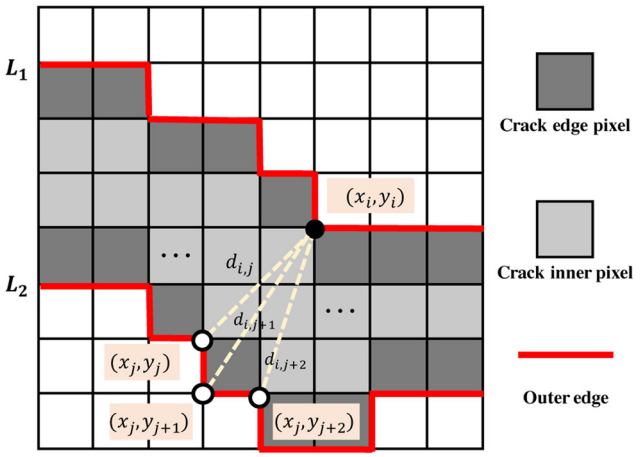


FIGURE 8 Crack width computation of point (x_i, y_i) on L_1

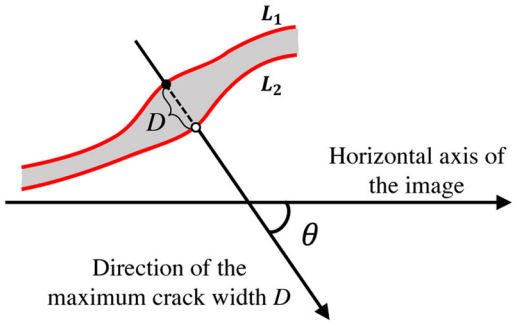


FIGURE 9 Maximum crack width error parameters

scaling factor, λ, converts the pixel value to crack width d_i (e.g., in [mm]). Finally, the maximum crack width, D, is obtained from all d_i, Equation (14).

$$d_i = \min_{(x_j, y_j) \in L_2} \left\langle (x_i, y_i), (x_j, y_j) \right\rangle \cdot \lambda \tag{13}$$

$$D = \max(d_i) \tag{14}$$

The theoretical maximum width error, err_θ, is computed, which provides an uncertainty range for the predicted maximum width D, that is, $[D - err_\theta, D + err_\theta]$. In addition, if the absolute value $|err_\theta|$ is larger than the predefined threshold value based on the demand, for example, the crack width limit is 0.20 mm in the Chinese Code (Ministry of transport of the People's Republic of China, 2018), this implies that the predicted crack width has high uncertainty and may not be reliable. This error is mainly attributed to the approximation of the pixels in the direction of the crack width. The projection angle, θ, is defined by the direction of the maximum crack width, D, and the horizontal axis of the image, Figure 9. The ranges of the theoretical coordinate differences after rounding in the

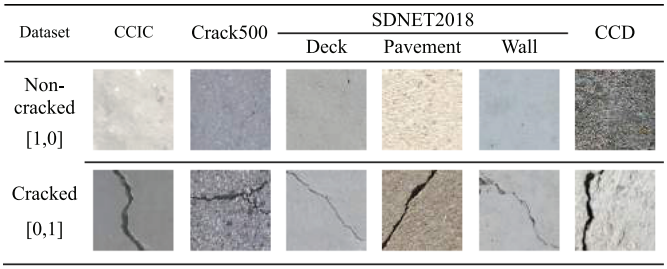


FIGURE 10 Samples from the source data set

horizontal, Δx, and vertical, Δy, axes are as follows:

$$\Delta x = D\cos\theta, \quad \Delta y = D\sin\theta, \quad 0° \le \theta \le 90° \tag{15}$$

$$\Delta x - 0.5\lambda < x < \Delta x + 0.5\lambda, \quad \Delta y - 0.5\lambda < y < \Delta y + 0.5\lambda \tag{16}$$

The theoretical error, err_θ, is derived and expressed as follows:

$$err_\theta = \left| \sqrt{D^2 + \lambda D(\cos\theta + \sin\theta) + 0.5\lambda^2} - D \right| \tag{17}$$

If actual (measured) maximum crack width, D_{act}, is available, the recognition performance of the CAL-ISM framework can be evaluated by the *recognition errors*, $(D - D_{act}) \big/ D_{act}$.

4 | NUMERICAL EXPERIMENTS

4.1 | Step 1: Benchmark model pretraining

The CAL-ISM framework is first trained and tested following its four major steps, where pretraining the benchmark model is the starting point. Two data sets are prepared, source data set for benchmark model pretraining and custom target data set for model fine-tuning and adaptation. The images of the source data set were selected from four open-source data sets, that is, Concrete Crack Images for Classification (CCIC) (Özgenel, 2018), Crack500 (Zhang et al., 2016), SDNET2018 (for bridge decks, highway pavements, and walls) (Dorafshan et al., 2018), and Concrete Crack Detection (CCD) (Satyen, 2018). All data have been labeled with "one-hot encoding," that is, [0,1] for cracked and [1,0] for noncracked. Few sample images are shown in Figure 10. A data set of 3000 labeled images is selected where the ratio between cracked (CR) and noncracked (NCR) labels is 1:1. Subsequently, all images are rescaled to $256 \times 256 \times 3$, and split into training and test sets with a 5:1 ratio, that is, 2500 labeled images used to

TABLE 1 Pretraining recognition performance of seven candidate models

Network	Parameters (#)	Accuracy (%)	Precision (%)	Recall (%)	F_1 score
VGG-11	160M	87.6	86.7	88.8	0.877
VGG-16	165M	92.2	96.1	88.0	0.919
ResNet-18	13M	94.0	95.8	92.0	0.939
ResNet-26	19M	94.2	96.6	91.6	0.940
Inception-ResNet-18	28M	94.8	95.9	93.6	0.947
SE-ResNet-18	13M	94.8	97.5	92.0	0.947
SE-Inception-ResNet-18	28M	94.8	97.9	91.6	0.946

train the benchmark model, and the remaining 500 labeled images used for validating the performance.

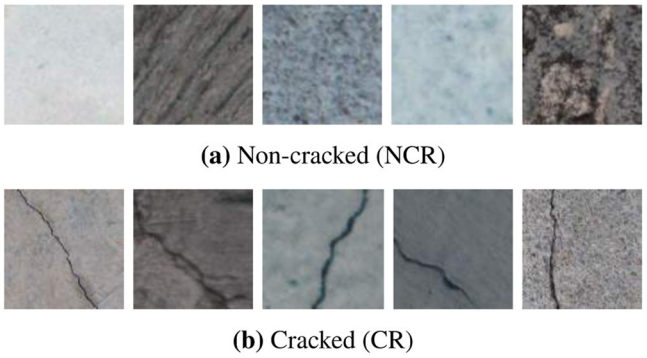
Multiple candidate DL structures are considered for a more robust performance. In this study, seven network structures, that is, VGG-11, VGG-16, ResNet-18, ResNet-26, Inception-ResNet-18, SE-ResNet-18, and SE-Inception-ResNet-18, are considered and compared. These seven structures are variants of several widely used network designs, and their model details can be found in He et al. (2016), Hu et al. (2018), Simonyan and Zisserman (2014), and Szegedy et al. (2017). Different from the original implementations, modifications are made to these models for (1) adjusting the number of convolutional layers and (2) matching the dimension of the *softmax* layer to the number of classes. Twenty images are processed per *batch* as dictated by the available computational capacity, and a total of 100 *epochs* (one epoch means looping over all data once) are applied, in which the optimization method to update the network parameters in back-propagation is the *stochastic gradient descent* (SGD). Through multiple runs for parameter tuning, the best performance is obtained for each candidate model, which reduces the uncertainty in the training procedure. The performance of all candidates with five metrics (number [#] of parameters, accuracy, precision, recall, and F_1, i.e., F_β, $\beta = 1$ due to a balanced 1:1 class ratio) are listed in Table 1. According to the results, using deeper network structure does not necessarily improve the model recognition performance, for example, compared to VGG-11 and ResNet-18, recall values of the VGG-16 and ResNet-26 slightly decrease. This is mainly attributed to the used small-scale data set for only two classes identified where the deeper network introduces more complexity and overfitting issues than the shallow one. From the comprehensive model performance evaluation for all seven models, SE-ResNet-18 has the *least* # of parameters (13 million [13M]) and *best* accuracy with highest F_1 score and second highest precision and recall. Thus, SE-ResNet-18 is selected as the benchmark model for the subsequent steps. The configuration of SE-ResNet-18 is listed in Table 2, where RGB is for Red, Green, & Blue, Conv stands for Convolution, for example, 7×7, 64 means 64 7×7 size Conv filters, and, for example, $\begin{bmatrix} 3 \times 3, 128 \\ 3 \times 3, 128 \end{bmatrix} \times 2$ represents repeating twice the two-layer structure using 128 3×3 Conv filters. More details can be found in Hu et al. (2018).

TABLE 2 Network configuration of the benchmark model

Layer	Output Size	Remarks
Input	256×256	Input RGB image
Conv block 1	128×128	7×7, 64, stride 2
Max pooling	64×64	3×3, stride 2
Conv block 2	64×64	$\begin{bmatrix} 3 \times 3, 64 \\ 3 \times 3, 64 \end{bmatrix} \times 2$
Conv block 3	32×32	$\begin{bmatrix} 3 \times 3, 128 \\ 3 \times 3, 128 \end{bmatrix} \times 2$
Conv block 4	16×16	$\begin{bmatrix} 3 \times 3, 256 \\ 3 \times 3, 256 \end{bmatrix} \times 2$
Conv block 5	8×8	$\begin{bmatrix} 3 \times 3, 512 \\ 3 \times 3, 512 \end{bmatrix} \times 2$
Global pooling	1,000	–
FC-layer	2	Activation: Softmax

4.2 | Step 2: Benchmark model retraining

The pretraining can provide better parameter initialization, and in this step, a new data set similar to the target domain, that is, bridge damage, is collected and partially labeled. To construct this target data set (Figure 11), a total of 175 images with high resolution of 6000 × 4000 are collected by the authors. This data set mainly consists of laboratory concrete components and campus bridges with both cracked (CR) and noncracked (NCR) classes. The acquisition equipment is Nikon D7100, and all images are vertical surfaces of the objects. Following a similar procedure to the one in Krizhevsky et al. (2012), data augmentation is applied to all images to increase the data set size. For a given image, a random 3000 × 3000 pixels region is cropped and then saved as a new image. This is repeated five times for each image in the data set. Subsequently, all images are rescaled to 256 × 256 pixels.



(a) Non-cracked (NCR)

(b) Cracked (CR)

FIGURE 11 Samples from the target data set

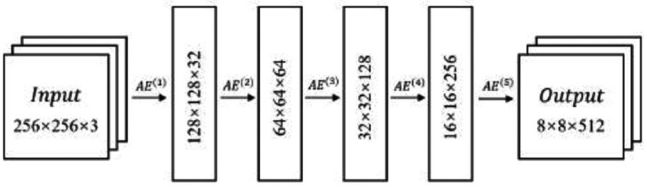


FIGURE 12 Network configuration of the SCAE

TABLE 3 Network configuration of the $AE^{(k)}$

Layer	$AE^{(1)}$	$AE^{(k)}$[a]
Input	$256 \times 256 \times 3$	$m \times m \times n$
Conv layer	3×3, 32, stride 2	3×3, $2n$, stride 2
Conv layer	1×1, 32, stride 1	1×1, $2n$, stride 1
Output	$128 \times 128 \times 32$	$m/2 \times m/2 \times 2n$

[a]For $k = 2, m = 128$ & $n = 32$, for $k = 3, m = 64$ & $n = 64$, for $k = 4, m = 32$ & $n = 128$, and for $k = 5, m = 16$ & $n = 256$.

For the purpose of validating the effectiveness of reducing labeling efforts, in this experimental part, no labeled training data are provided in advance, and the pretrained classifier starts from step 2 of the active learning procedure with unlabeled data. Based on this setting, the target data set is established, which consists of unlabeled training set, *Unl*, with 60,000 images, and labeled test set, L, with 1000 images, where 176 and 824 images are labeled as CR and NCR, respectively, that is, L is moderately imbalanced with a NCR:CR ratio of about 4.7. Accordingly, the labeling operations are only conducted for fewer images during each training iteration.

The configuration of the SCAE is illustrated in Figure 12, and details about each AE layer is listed in Table 3. In order to avoid both gradient explosion and gradient vanishing issues, features are extracted layer by layer, and normalization is performed before feature stacking. From the procedure of the SCAE-based active learning, the unlabeled data, *Unl*, are separated into $M = 6$ batches. Furthermore, for simplicity, $N_d = 100$ most difficult images are selected for manual labeling in each iteration. Unlike the source data set with a balanced 1:1 class ratio used in Step 1, the target data set has a moderately imbalanced class ratio, that is, 4.7. Thus, $\beta = 2$ is considered for F_β score evaluation, that is, F_2, similar to (Gao et al., 2021).

After six rounds of iterations, the retrained benchmark model using SCAE-based active learning (SCAE-AL) obtained a stable and satisfying performance, as shown in Figure 13. In addition, comparisons are made with other methods, that is, self-training (Self-T) and active learning (AL), where results are summarized in Table 4. Even though the benchmark obtained 90.0% accuracy on the target data set, this is deceptive. Due to the skewed class-distribution of the target data set, the recall (TPR) value of the benchmark is just slightly higher than 60%. Considering its high TNR value (95.6%), the results indicate that the benchmark is more prone to predicting the images as noncracked. In addition, Self-T and AL models work better than the benchmark, where AL is slightly better than Self-T because of the assistance of human-corrected labels. However, the improvements for each metrics are limited (less than 5%). Considering that both Self-T and AL need limited labeling efforts, that is, not required for Self-T and only 600 images are manually relabeled among 60,000 unlabeled data for AL, such a small improvement is still acceptable.

Adding the SCAE into the AL, the SCAE-AL model obtained significant improvements in both precision (from 75.7% to 84.1%) and recall (from 63.6% to 90.3%), which relieve the previously mentioned bias issue. This is explained by a stronger feature representation ability through feature fusion from the benchmark model and the SCAE structure. The semisupervised mechanism in the SCAE branch improves the utilization of the abundant unlabeled data in the target data set, which makes the SCAE able to extract deep features from the target data set, and thus improves the generalization ability of the SCAE. Furthermore, through feature fusion, extracted features from the benchmark and the SCAE are combined, providing more comprehensive information about the target data set. Therefore, the SCAE-AL achieved a better generalization ability than the other methods.

4.3 | Step 3: Crack segmentation

In step 2, the benchmark model is updated on the target data set using SCAE-based active learning, and it can be directly used as a crack detector for images acquired from real-world projects. Subsequently, the detected "cracked" images are further processed by the segmentation model for mask generation in step 3.

Similar to the pretraining step, three candidate segmentation models, that is, FCN, PSPNet, and MR-CNN

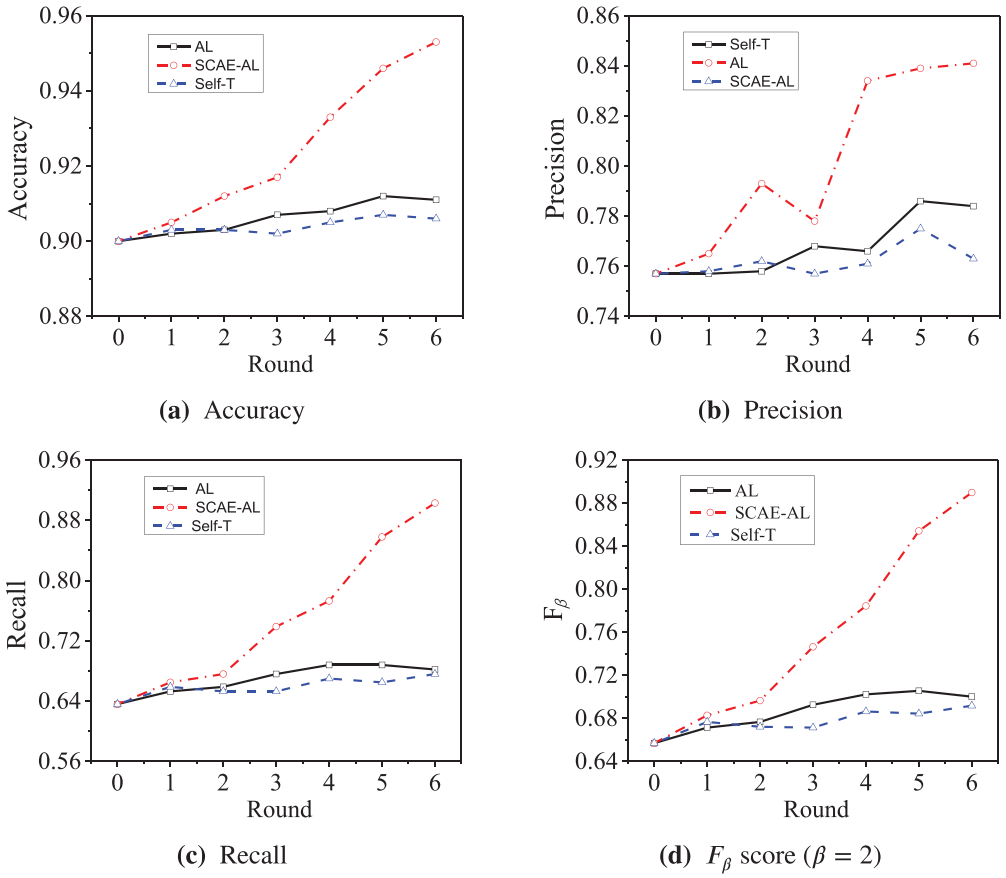


FIGURE 13 Evaluation of the model retraining

TABLE 4 Retraining recognition performance of the benchmark model on the target data set

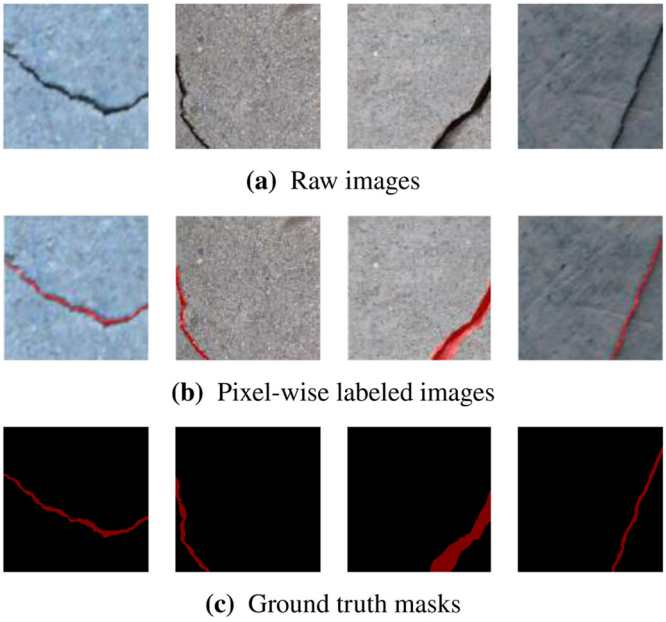
Network	Accuracy (%)	Precision (%)	Recall (%)	F_2 score	TNR (%)
Benchmark	90.0	75.7	63.6	0.657	95.6
Self-T	90.6	76.3	67.6	0.692	95.5
AL	91.1	78.4	68.2	0.700	96.0
SCAE-AL	95.3	84.1	90.3	0.890	96.4

are considered in the training for robustness, where the best model with the highest MIoU is selected as the final segmentation model for practical application. Based on multiple trials, different feature extraction networks are used in each segmentation model, that is, VGG-16 (Simonyan & Zisserman, 2014) for FCN, ResNet101 (He et al., 2016) for PSPNet, and ResNeXt101-FPN (He et al., 2017) for MR-CNN. In addition, all three models can receive arbitrary size of the input images. To train and validate these candidate models, another pixel-level labeled data set is required. Thus, additional 244 "cracked" images are collected and labeled pixel-wise using the open-source tool *Labelme* (Wada, 2016). Examples from this data set, used for segmentation, are shown in Figure 14.

After 100 training epochs, all three candidate models stabilized with flat loss history curves. The obtained best MIoU results for FCN, PSPNet, and MR-CNN are 0.71, 0.77, and 0.81, respectively. Based on visualization of the segmentation on random sample images, Figure 15, MR-CNN has more accurate segmentation performance, higher MIoU, closer to the ground truth mask, and lower noise observed in the mask. Therefore, MR-CNN model is recommended as the segmentation model of choice for real-time BHM projects.

4.4 | Step 4: Crack width measurement

In step 4, the generated mask is further processed to extract edges of the crack through the modified Sobel operator and the 8-direction region growing algorithm. Three samples are presented in Figure 16, where Figure 16a includes a

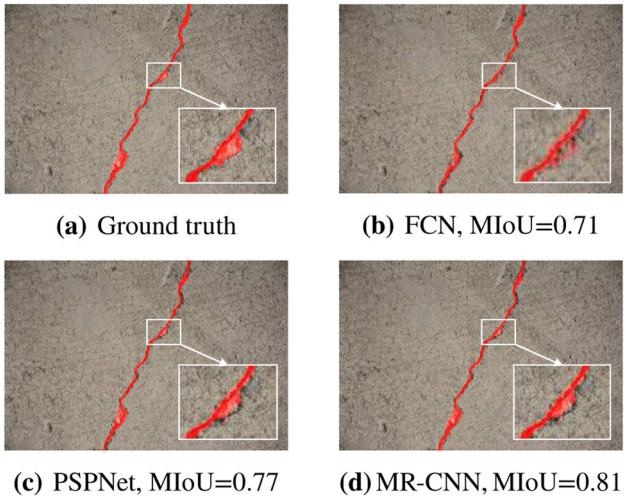


(a) Raw images

(b) Pixel-wise labeled images

(c) Ground truth masks

FIGURE 14 Samples from the data set used for segmentation



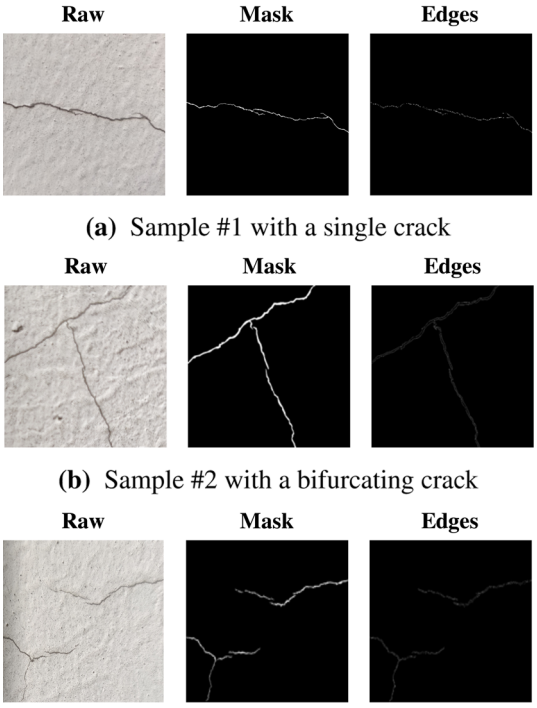
(a) Ground truth (b) FCN, MIoU=0.71

(c) PSPNet, MIoU=0.77 (d) MR-CNN, MIoU=0.81

FIGURE 15 Examples of segmentation performance

single crack as a simple scenario, Figure 16b has a bifurcating crack as a more complex scenario, and Figure 16b includes multiple irregular cracks as the most complex scenario. The actual crack widths of these samples are measured by the crack width measurement instrument ZBL-F103 developed by Beijing Zhibolian Technology Co., Ltd., and the instrument accuracy is about 0.01 mm. Based on measurements and calibration during data collection, the scale factor λ for each sample image is computed from the ratio of actual length and corresponding number of pixels.

According to Equations (11) to (17), the predicted crack width and related information are listed in Table 5. Comparing the predicted maximum crack width, D, to the



(a) Sample #1 with a single crack

(b) Sample #2 with a bifurcating crack

(c) Sample #3 with multiple cracks

FIGURE 16 Measurements of three types of crack images

actual (measured) crack width, D_{act}, the *recognition errors* are also computed. Based on codes of China (Ministry of transport of the People's Republic of China, 2018) and other countries, the crack width limit is typically 0.20 mm. In this experiment, the CAL-ISM framework recognizes multiple types of cracks, including cracks less than 0.20 mm in width. In addition, the average error is 3% to 4%, satisfying practical engineering accuracy.

5 | TEST SPECIMEN APPLICATION

In this section, a quasi-statically tested RC bridge pier specimen, Figure 17, is selected as an application to evaluate the proposed CAL-ISM framework for crack detection and measurement. The net height of the pier is 1800 mm measured from the top of the footing to the bottom of the bent cap. The cross-section of the pier is a square (500×500 mm^2). The west side (marked "W" in Figure 17a,b), with a rectangle area of $500 \times 1,300$ mm^2, of the pier is selected for crack identification and measurement. This side is divided into 65 subblocks (i.e., 100×100 mm^2 each) and numbered from 1 to 65, as shown in Figure 17b,c. Each subblock is photographed by a high-resolution camera of 24 megapixels. These images are somewhat irregular quadrilaterals because the camera could not be perpendicularly mounted to the concrete surface of the pier

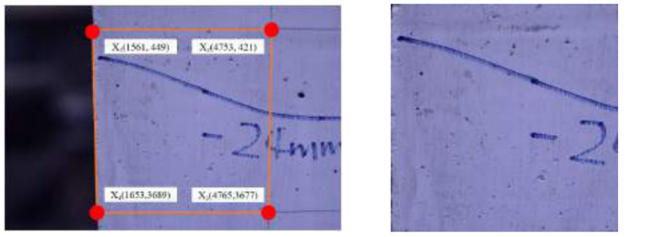
TABLE 5 Crack width measurement of three sample images

Sample #	Max crack width $D \pm err_\theta$ (mm)	Angle θ (°)	Actual crack width (mm)	Recognition error (%)
1	0.134 ± 0.011	60.25	0.130	3.08
2	0.145 ± 0.012	49.39	0.140	3.57
3	0.186 ±0 .012	29.05	0.180	3.33



(a) RC bridge pier specimen (shown in the foreground) **(b)** Marking mesh of surface **(c)** Sub-block divisions

FIGURE 17 Test specimen and its subblock divisions



(a) Quadrilateral sub-block **(b)** Processed squared image

FIGURE 18 Transforming quadrilateral subblock to square

during photographing. Thus, a coordinates transformation algorithm is employed to convert these quadrilaterals to squares. Figure 18a is a typical quadrilateral image of a subblock and Figure 18b is a transformed square image with a resolution of 4000 × 4000 pixels. Thus, the pixel width is $\lambda = 100/4000 = 0.025$ mm/pix using the subblock size.

A large number of cracks are irregularly distributed on the surface of the RC pier and concrete surface spalling mostly occurs near the bottom of the pier. To distinguish cracked, spalled, or intact concrete on the surface of the pier and to conveniently measure the widths of the cracks, three colored pixels (i.e., black, red, and green) corresponding to the three damage states, that is, nondamage (intact), cracked, and spalled, respectively, are considered. Only the parts on the surface marked in the red channel, representing cracks, are measured.

Since DL models in CAL-ISM have been trained, the images containing cracks are directly fed to the classification model to identify the existence of cracks. Subsequently, the predicted "cracked" images are processed by the segmentation model to generate masks, which highlight the pixels of the cracks and the lines marked using marking pens. Based on the mask images, the crack width is measured. In this application, detection is conducted for all subblocks following the CAL-ISM procedure, and the results for all subblocks are assembled into a whole picture using their numbers (1 to 65), Figure 19. For better visualization, nondamage areas, cracks, and spalling zones on the surface of the pier are recolored in white, black, and purple, respectively. From Figure 19, all major cracks are identified. Subsequently, as an example of crack width measurements, one region concatenated from 10 subblocks (46 to 55), is selected, Figure 20. Each crack is recognized by CAL-ISM and compared with the actual measured width, Table 6.

The obtained results indicate that the identified crack width is always conservatively larger than the measured value. This is attributed to the existence of surface defects, for example, small pits, on the crack path, making it difficult for the DL model to distinguish these defects. On the contrary, such defects can be avoided in actual measurements. From Table 6, most errors of the recognized crack width are less than 10%, especially for actual crack width less than 0.5 mm (an exception is crack # 4). For

TABLE 6 Crack width identification of the pier specimen

Crack #	Max width $D \pm err_\theta$ (mm)	Angle θ (°)	Measured width (mm)	Recognition error (%)
1	0.833 ± 0.012	46.51	0.82	1.6
2	0.178 ± 0.011	31.12	0.17	4.7
3	0.415 ± 0.011	58.05	0.39	6.4
4	0.226 ± 0.008	89.48	0.20	13.0
5	0.864 ± 0.011	28.16	0.81	6.7
6	0.449 ± 0.012	43.60	0.41	9.5
7	0.247 ± 0.008	0.73	0.23	7.4
8	1.313 ± 0.010	69.88	0.97	35.4
9	0.974 ± 0.011	61.61	0.76	28.2
10	1.798 ± 0.011	53.24	1.66	8.3

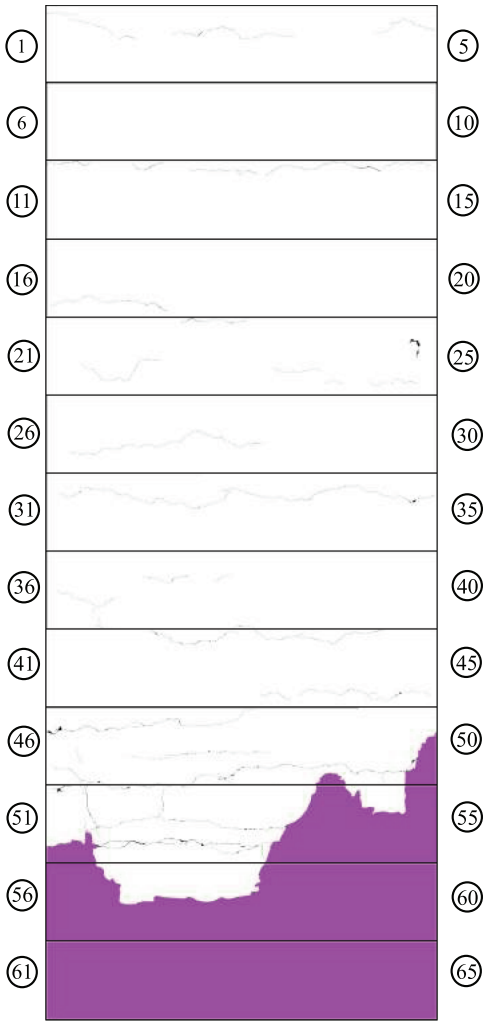


FIGURE 19 Crack identification of the tested pier

some wide cracks (# 8 & 9), due to the negative effects of surface defects, the identified crack width is overestimated with larger errors (≈ 28% to 35%). In conclusion, the comparison between identified and measured crack widths indicates that the proposed CAL-ISM framework is accurate, especially for narrow cracks, and suitable for BHM applications.

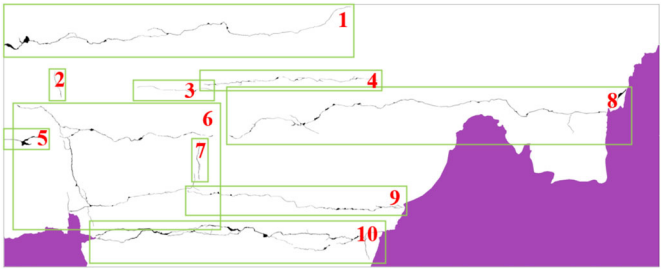


FIGURE 20 Extracted 10 cracks in subblocks 46 to 55

6 | BRIDGE FIELD APPLICATION

To demonstrate the proposed vision-based CAL-ISM framework for crack detection and measurement of real-world BHM projects, two RC bridge field applications, namely, Wutang bridge and Xiejinghe bridge, are studied.

6.1 | Wutang bridge

Wutang bridge is a three-span (10, 13, and 10 m) simply supported bridge located in Shanghai, China (N31.3745, E121.1454), Figure 21a. A surface area (400×500 mm^2) located at the bottom of the hollow slab girder near the abutment having minor cracks is selected for investigation. This selected area is divided into 20 square subblocks (100×100 mm^2) and photographed with 20 images composed of 2560×2560 pixels each. In other words, $\lambda = 100/2560 = 0.0391$ mm/pix. Subsequently, these 20 images are processed by a quadrilateral transformation to convert them to square subblocks. Finally, they are spliced together to restore back the investigated surface area.



FIGURE 21 BHM projects of Wutang bridge and Xiejinghe bridge, Shanghai, China

TABLE 7 Crack width identification of Wutang bridge

Crack #	Max width $D \pm err_\theta$ (mm)	Angle θ (°)	Measured width (mm)	Recognition error (%)
1	0.200 ± 0.020	0.00	0.18	11.1
2	0.180 ± 0.027	33.69	0.17	5.9
3	0.825 ± 0.024	14.03	0.80	3.1
4	0.861 ± 0.021	2.73	0.77	12.2
5	0.403 ± 0.022	82.87	0.40	0.8
6	0.762 ± 0.026	23.20	0.72	5.8
7	0.570 ± 0.027	52.13	0.55	3.6

The trained recognition and segmentation models in CAL-ISM are employed to identify and measure the cracks in the designated region of the bridge surface, and then the recognized cracks are exhibited and numbered in Figure 22a. The models accurately identify and segment the cracks. Note that cracks 5, 6, and 7 belong to three parts of the same crack. However, several stains on the surface partly covered the crack leading to identifying it as three discrete segments. The estimated widths of the cracks are listed in Table 7 along with the actual width measurements. From the results, the estimated crack widths are conservatively larger than the measured ones. Additionally, the recognition errors are less than 13%, and some of them are even less than 5%. Such estimation accuracy is judged to be acceptable in BHM practice.

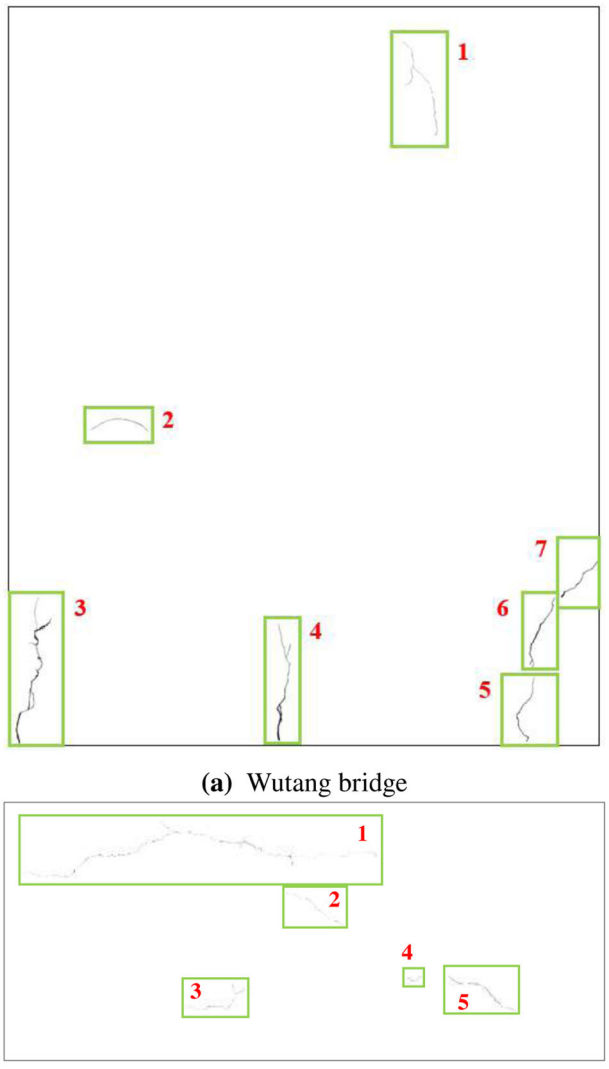
The detection results of the maximum crack width located in different regions of the Wutang bridge show that five (3 to 7) from the seven cracks are larger than 0.2 mm. Consequently, according to most codes (e.g., Ministry of transport of the People's Republic of China, 2018), the bridge should be repaired by injection to enhance its durability by eliminating deterioration due to water penetration, corrosion, and stress concentration caused by these wide cracks. Although only a small surface region of the bridge is studied as an example for demonstrating the proposed vision-based CAL-ISM framework, the whole surface of the bridge, especially the surfaces of the damage-critical parts in the bridge, for example, the piers, can be examined using the same framework.

6.2 | Xiejinghe bridge

Built in 2001, Xiejinghe bridge is a three-span (9 m/span) simply supported bridge, which is located in Shanghai, China (N31.36847, E121.12453), as shown in Figure 21b. A surface area (300×700 mm^2) at the bottom of the RC girder is selected for crack investigation where there are several cracks with different widths. This area is divided into 21 subblocks (100×100 mm^2) and photographed with 21 images composed of 2560×2560 pixels each. Subsequently, these 21 images are processed by a quadrilateral

TABLE 8 Crack width identification of the Xiejinghe bridge

Crack #	Max width $D \pm err_\theta$ (mm)	Angle θ (°)	Measured width (mm)	Recognition error (%)
1	0.141 ±0.027	36.70	0.14	0.7
2	0.100 ± 0.028	44.20	0.10	0.0
3	0.112 ± 0.021	90.00	0.11	1.8
4	0.112 ±0.026	67.50	0.11	1.8
5	0.123 ± 0.022	86.60	0.12	2.5



(a) Wutang bridge

(b) Xiejinghe bridge

FIGURE 22 Crack identification of two bridges

transformation to convert them to square subblocks. Finally, they are spliced together to restore back the investigated surface area, Figure 21b.

Similar to the Wutang bridge, CAL-ISM is applied to the Xiejinghe bridge and the resulting cracks and their numbering are shown in Figure 22b. A total of five cracks are finally identified within the selected area and their maximum widths together with the measured ones are all listed in Table 8. It indicates that the identified maximum widths match well the measured ones, where the maximum recognition error is only 2.5%. This reconfirms that the CAL-ISM framework can be used to not only identify and segment cracks but also to accurately predict their widths in practical scenarios. Moreover, all maximum widths of the detected cracks are less than 0.2 mm. Consequently, the present health condition of the Xiejinghe bridge is "good" considering usual maintenance principles in most codes (e.g., Ministry of transport of the People's Republic of China, 2018). However, since the bridge is more than 20 years old, it needs to be well-maintained in future service life.

In summary, the applications of the CAL-ISM framework for crack detection and measurement for bridges under service conditions and following earthquake events are successful. It is expected that the CAL-ISM framework will initiate new directions for effective BHM with a strong positive impact on future bridge maintenance and reconnaissance efforts.

7 | CONCLUSIONS AND EXTENSIONS

In this paper, we propose a multistep crack recognition, segmentation, and width measurement framework, namely, "CAL-ISM." It consists of four steps: (1) pretraining benchmark classification model, (2) retraining semisupervised active learning model, (3) pixel-level crack segmentation, and (4) crack width measurement. In addition, CAL-ISM is applied and validated through practical applications: (1) bridge column test specimen in the laboratory and (2) bridge structure monitoring in the field. The results demonstrate the effectiveness and efficiency of the CAL-ISM for BHM applications. Key conclusions are presented as follows:

(1) Retraining by semisupervised active learning, with limited labeling efforts (only labeled 600 among 60,000 images), the benchmark classification model achieves better generalization ability with significant accuracy improvement, for example, F_2 score,

precision, and recall increased by 23.3%, 8.4%, and 26.7%, respectively.

(2) Based on the recognized "cracked" images from the retrained classification model, the MR-CNN achieves the best MIoU, that is, 0.81, among all models, and is selected as the segmentation model for mask extraction.

(3) Based on the developed modified Sobel operator, the edges of the crack are successfully extracted from the mask so that the outlines of the crack are obtained for the crack width measurements.

(4) In the application of the tested bridge column specimen, the identification results demonstrate a satisfactory accuracy of the CAL-ISM framework for the various forms of cracks on the column. The obtained maximum recognition error for most narrow cracks (less than 0.2 mm) is limited to less than 10%.

(5) In the application of the BHM for two actual bridge projects, the cracks on the bottom of the bridge girder are identified. The results are conservative compared with measured values and the error range is about 0 to 12%, which is acceptable for practical BHM.

Several topics needing future investigation are as follows:

(1) To further enhance the accuracy of the classification performance, more powerful methods, such as Enhanced Probabilistic Neural Network (Ahmadlou & Adeli, 2010), Neural Dynamic Classification algorithm (Rafiei & Adeli, 2017), Dynamic Ensemble Learning algorithm (Alam et al., 2020), Finite Element Machine for fast learning (Pereira et al., 2020), and BSS-GAN (Gao et al., 2021), can replace the backbone network of the CAL-ISM, and more comparative studies need to be conducted.

(2) The active learning setting in CAL-ISM reduces the tedious manual labeling efforts. However, currently, it only works for classification models in the first two steps, and it cannot contribute to the training of the segmentation labels in step 3, where the segmentation labeling is even more costly. Future explorations to reduce the segmentation labeling effort is recommended.

(3) In this paper, only cracks were discussed, but the actual bridge damage is very complex. Therefore, more images containing various damage patterns should be collected from various structural forms and locations to enrich the data set and improve model recognition performance.

(4) The CAL-ISM is suitable for other types of structures than bridges, for example, buildings, and its performance may vary due to using different sources of data and different amounts of manually labeled data. Therefore, more case studies are recommended for further explorations.

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